



Assignment 06 – Strings in Python

 Duration: 20–25 minutes

 Objective:

To strengthen your understanding of **indexing, slicing, and string methods** in Python.



1 Accessing Characters

Write a Python program that:

- Takes a string input from the user.
- Displays:
 1. The first and last character
 2. The third character from the end
 3. Every second character in the string

Example:

```
Enter a word: Programming
First: P
Last: g
Third from end: n
Every second character: Pormig
```



2 String Slicing

Write a program to:

- Take a word as input.
- Print the first half and the second half of the word separately.

Hint: Use slicing (`len(word)//2`) to find the midpoint.

Example:

```
Enter a word: Python
First half: Pyt
Second half: hon
```



3 String Methods

Take a sentence as input and perform the following:

- Convert it to uppercase and display it
- Count how many times the word "the" appears (use `.count()`)
- Replace "Python" with "AI"

Example:

```
Enter a sentence: Python is the best. The world loves Python!
UPPERCASE: PYTHON IS THE BEST. THE WORLD LOVES PYTHON!
Count of 'the': 2
Replaced: AI is the best. The world loves AI!
```



String Analysis

Write a program that takes a string and:

- Checks if it's all alphabets (`isalpha()`)
- Checks if it's all digits (`isdigit()`)
- Checks if it's alphanumeric (`isalnum()`)

Example:

```
Enter a value: Python3
Contains only alphabets? False
Contains only digits? False
Contains letters and numbers? True
```



5 Mini Project: Word Counter (Enhanced)

Extend the earlier **Word Counter** project:

- Input a sentence
- Remove punctuation (optional challenge: use `.replace()` or `strip()`)
- Count and display how many times each word appears
- Show the **most frequent word** at the end

Example:

```
Enter a sentence: Python is easy, and Python is fun!
Word counts:
python : 2
is : 2
easy : 1
and : 1
```

```
fun : 1
```

```
Most frequent word: python
```

🎲 Submission Tips

✅ Test each program with multiple inputs ✅ Use clear variable names and comments ✅ Display output in a readable format